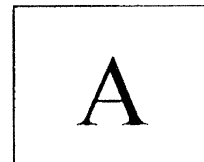


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**B.Tech. Degree V Semester Supplementary Examination**  
**April 2018**

CE/CS/EC/EE/IT/ME/SE  
AS 15-1501 NUMERICAL AND STATISTICAL METHODS  
(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

**PART A**  
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Prove that  $\Delta \log f(x) = \log \left( 1 + \frac{\Delta f(x)}{f(x)} \right)$ .
- (b) Evaluate  $\int_0^1 dx / (1 + x^2)$  using Simpson's 1/3 rule taking  $h = 1/4$ .
- (c) Derive a formula to find a real root of the equation  $f(x) = 0$  by Regula-Falsi method.
- (d) Solve  $y' = y - \frac{2x}{y}$ ,  $y(0) = 1$  in the range  $0 \leq x \leq 0.2$  using Euler's method. Take  $h = 0.1$ .
- (e) Find the mean and variance of a random variable  $X$  with probability density function  $f(x) = \frac{3}{2}(1 - x^2)$ ,  $0 < x < 1$ .
- (f) Write down the probability density function of the binomial distribution for which mean is 4 and variance 3.
- (g) Define : (i) Level of Significance (ii) Power of the test.
- (h) The sizes and mean of 2 independent random samples are 400, 225, 3.5 and 3 respectively. Can we conclude that the samples are drawn from the same population with SD 1.5?
- (i) If  $X$  follows Poisson distribution with  $P(X = 2) = \frac{2}{3} P(X = 1)$ , then find  $P(X = 3)$ .
- (j) A random sample of size 18 is taken from a Normal distribution. Test the hypothesis  $H_0 : \sigma^2 = 0.36$  against  $H_1 : \sigma^2 > 0.36$  given  $S^2 = 0.68$ ,  $\alpha = 5\%$ .



**PART B**

(4 × 10 = 40)

- II. (a) Apply Gauss – Seidal iteration method to solve the equations  
 $20x + y - 2z = 17$ ,  $3x + 20y - z = -18$ ,  $2x - 3y + 20z = 25$ .
- (b) Use Newton's divided difference formula to find  $f(7)$  if  $f(3) = 24$ ,  $f(5) = 120$ ,  $f(8) = 504$ ,  $f(9) = 720$  and  $f(12) = 1716$ .

**OR**

(P.T.O.)

- III. (a) Given the values

|        |    |   |    |     |
|--------|----|---|----|-----|
| $x$    | 0  | 2 | 3  | 6   |
| $f(x)$ | -4 | 2 | 14 | 158 |

Using Lagrange's formula for interpolation, find the value of  $f(4)$ .

- (b) From the following table of half yearly premium for policies maturing at different ages, estimate the premium for policies maturing at ages 46 and 63.

|                 |      |     |     |     |     |
|-----------------|------|-----|-----|-----|-----|
| Age( $x$ )      | 45   | 50  | 55  | 60  | 65  |
| Premium ( $y$ ) | 1150 | 960 | 830 | 740 | 680 |

- IV. (a) Obtain the first and second derivatives of  $\sqrt{x}$  at  $x = 15$  from the table.

|            |       |       |       |       |       |       |
|------------|-------|-------|-------|-------|-------|-------|
| ( $x$ )    | 15    | 17    | 19    | 21    | 23    | 25    |
| $\sqrt{x}$ | 3.873 | 4.123 | 4.359 | 4.583 | 4.796 | 5.000 |

- (b) The velocity  $v(km/min)$  of a moped which starts from rest is given at fixed intervals of time  $t(min)$  as follows :

|       |    |    |    |    |    |    |    |    |    |    |
|-------|----|----|----|----|----|----|----|----|----|----|
| $t :$ | 2  | 4  | 6  | 8  | 10 | 12 | 14 | 16 | 18 | 20 |
| $v :$ | 10 | 18 | 25 | 29 | 32 | 20 | 11 | 5  | 2  | 0  |

Estimate approximately the distance covered in 20 minutes.

OR

- V. (a) Solve  $y' = 3x + \frac{y}{2}$  and  $y(0) = 1$  by Taylor's series method and compute  $y(0.1)$  and  $y(0.2)$ .

- (b) Apply Runge-Kutta method of fourth order to find an approximate value of  $y$  when  $x = 0.2$  given that  $dy/dx = x + y$  and  $y = 1$  when  $x = 0$ .

- VI. (a) Derive the mean and variance of Binomial distribution.

- (b) Fit a Poisson distribution to the following data and test the goodness of fit.

|          |     |    |    |   |   |
|----------|-----|----|----|---|---|
| $x :$    | 0   | 1  | 2  | 3 | 4 |
| $f(x) :$ | 109 | 65 | 22 | 3 | 1 |

OR

- VII. (a) Fit a curve of the form  $y = ae^{bx}$  to the following data by the method of least squares.

|       |     |     |      |      |     |     |
|-------|-----|-----|------|------|-----|-----|
| $x :$ | 1   | 2   | 3    | 4    | 5   | 6   |
| $y :$ | 1.6 | 4.5 | 13.8 | 40.2 | 125 | 300 |

- (b) The mean mark on a final examination was 72 and Standard deviation 9. The top 10% of the students are to receive Grade A. What is the minimum mark a student must get in order to receive Grade A, if the marks are normally distributed?

(Contd...3)

- VIII. (a) A population follows a normal distribution with mean  $\mu$  and variance 9. To test  $H_0: \mu = 5$  against  $\mu > 7$ , the test procedure is to reject  $H_0$  if  $\bar{x} \geq 6$  where  $\bar{x}$  is the mean of a sample of size 16. Find the significance level and power of the test.
- (b) Two independent random samples of sizes  $n_1 = 10$ ,  $n_2 = 7$  were observed to have sample variance  $s_1^2 = 16$ ,  $s_2^2 = 3$ . Using  $\alpha = 0.1$ , test  $H_0: \sigma_1^2 = \sigma_2^2$  against  $H_1: \sigma_1^2 \neq \sigma_2^2$ .

OR

- IX. (a) A company claims that its light bulbs are superior to those of its main competitor. A study shows that a sample of size  $n_1 = 40$  of its bulbs has a mean life time of continuous use with a SD of 27 hours, while a sample of size  $n_2 = 40$  bulbs made by its main competitor has a mean life time of 638 hours of continuous use with a SD of 31 hours. Can we accept the claim of the company at 0.05 level of significance?
- (b) A random sample of size 12 is taken from a normal population with mean  $\mu$  and variance 9. Find the probability that the variance of the sample lies between 3.4 and 14.8.

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